GENERATIVE MODELS AND MODEL CRITICISM VIA OPTIMIZED MAXIMUM MEAN DISCREPANCY

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ABSTRACT

We propose a method to optimize the representation and distinguishability of samples from two probability distributions, by maximizing the estimated power of a statistical test based on the maximum mean discrepancy (MMD). This optimized MMD is applied to the setting of unsupervised learning by generative adversarial networks (GAN), in which a model attempts to generate realistic samples, and a discriminator attempts to tell these apart from data samples. In this context, the MMD may be used in two roles: first, as a discriminator, either directly on the samples, or on features of the samples. Second, the MMD can be used to evaluate the performance of a generative model, by testing the model's samples against a reference data set. In the latter role, the optimized MMD is particularly helpful, as it gives an interpretable indication of how the model and data distributions differ, even in cases where individual model samples are not easily distinguished either by eye or by classifier.

This post-publication revision corrects some errors in constants of the estimator (5). *The appendix deriving the estimator has been replaced by* Sutherland *(*2019*).*

1 INTRODUCTION

Many problems in testing and learning require evaluating distribution similarity in high dimensions, and on structured data such as images or audio. When a complex generative model is learned, it is necessary to provide feedback on the quality of the samples produced. The generative adversarial network (Goodfellow et al., 2014; Gutmann et al., 2014) is a popular method for training generative models, where a rival discriminator attempts to distinguish model samples from reference data. Training of the generator and discriminator is interleaved, such that a saddle point is eventually reached in the joint loss.

A useful insight into the behavior of GANs is to note that when the discriminator is properly trained, the generator is tasked with minimizing the Jensen-Shannon divergence measure between the model and data distributions. When the model is insufficiently powerful to perfectly simulate the test data, as in most nontrivial settings, the choice of divergence measure is especially crucial: it determines which compromises will be made. A range of adversarial divergences were proposed by Huszar (2015), using a weight to interpolate between KL, inverse KL, and Jensen-Shannon. This weight may be interpreted as a prior probability of observing samples from the model or the real world: when there is a greater probability of model samples, we approach reverse KL and the model seeks out modes of the data distribution. When there is a greater probability of drawing from the data distribution, the model approaches the KL divergence, and tries to cover the full support of the data, at the expense of producing some samples in low probability regions.

This insight was further developed by Nowozin et al. (2016), who showed that a much broader range of f-divergences can be learned for the discriminator in adversarial models, based on the variational formulation of f-divergences of Nguyen et al. (2008). For a given f-divergence, the model learns the composition of the density ratio (of data to model density) with the derivative of f, by comparing generator and data samples. This provides a lower bound on the "true" divergence that would be obtained if the density ratio were perfectly known. In the event that the model is in a smaller class than the true data distribution, this broader family of divergences implements a variety of different approximations: some focus on individual modes of the true sample density, others try to cover the support. It is straightforward to visualize these properties in one or two dimensions (Nowozin et al., 2016, Figure 5), but in higher dimensions it becomes difficult to anticipate or visualize the behavior of these various divergences.

An alternative family of divergences are the integral probability metrics (Müller, 1997), which find a witness function to distinguish samples from P and Q.[1] A popular such class of witness functions in GANs is the maximum mean discrepancy (Gretton et al., 2012a), simultaneously proposed by Dziugaite et al. (2015) and Li et al. (2015). The architecture used in these two approaches is actually quite different: Dziugaite et al. use the MMD as a discriminator directly at the level of the generated and test images, whereas Li et al. apply the MMD on input features learned from an autoencoder, and share the decoding layers of the autoencoder with the generator network (see their Figure 1(b)). The generated samples have better visual quality in the latter method, but it becomes difficult to analyze and interpret the algorithm given the interplay between the generator and discriminator networks. In a related approach, Salimans et al. (2016) propose to use feature matching, where the generator is tasked with minimizing the squared distance between expected discriminator features under the model and data distributions, thus retaining the adversarial setting.

In light of these varied approaches to discriminator training, it is important to be able to evaluate quality of samples from a generator against reference data. An approach used in several studies is to obtain a Parzen window estimate of the density and compute the log-likelhiood (Goodfellow et al., 2014; Nowozin et al., 2016; Breuleux et al., 2011). Unfortunately, density estimates in such high dimensions are known to be very unreliable both in theory (Wasserman, 2006, Ch. 6) and in practice (Theis et al., 2016). We can instead ask humans to evaluate the generated images (Denton et al., 2015; Salimans et al., 2016), but while evaluators should be able to distinguish cases where the samples are over-dispersed (support of the model is too large), it may be more difficult to find under-dispersed samples (too concentrated at the modes), or imbalances in the proportions of different shapes, since the samples themselves will be plausible images. Recall that different divergence measures result in different degrees of mode-seeking: if we rely on human evaluation, we may tend towards always using divergences with under-dispersed samples.

We propose to use the MMD to distinguish generator and reference data, with features and kernels chosen to maximize the test power of the quadratic-time MMD of Gretton et al. (2012a). Optimizing MMD test power requires a sophisticated treatment due to the different form of the null and alternative distributions (Section 2). We also develop an efficient approach to obtaining quantiles of the MMD distribution under the null (Section 3). We demonstrate on simple artificial data that simply maximizing the MMD (as in Sriperumbudur et al., 2009) provides a less powerful test than our approach of explicitly maximizing test power. Our procedure applies even when our definition of the MMD is computed on features of the inputs, since these can also be trained by power maximization.

When designing an optimized MMD test, we should choose a kernel family that allows us to visualize where the probability mass of the two samples differs most. In our experiments on GAN performance evaluation, we use an automatic relevance determination (ARD) kernel over the output dimensions, and learn which coordinates differ meaningfully by finding which kernels retain significant bandwidth when the test power is optimized. We may further apply the method of Lloyd & Ghahramani (2015, Section 5) to visualize the witness function associated with this MMD, by finding those model and data samples occurring at the maxima and minima of the witness function (i.e., the samples from one distribution least likely to be in high probability regions of the other). The optimized witness function gives a test with greater power than a standard RBF kernel, suggesting that the associated witness function peaks are an improved representation of where the distributions differ. We also propose a novel generative model based on the feature matching idea of Salimans et al. (2016), using MMD rather than their "minibatch discrimination" heuristic, for a more principled and more stable enforcement of sample diversity, without requiring labeled data.

[1] Only the total variation distance is both an f-divergence and an IPM (Sriperumbudur et al., 2012).

2 MAXIMIZING TEST POWER OF A QUADRATIC MMD TEST

Our methods rely on optimizing the power of a two-sample test over the choice of kernel. We first describe how to do this, then review alternative kernel selection approaches.

2.1 MMD AND TEST POWER

We will begin by reviewing the maximum mean discrepancy and its use in two-sample tests. Let k be the kernel of a reproducing kernel Hilbert space (RKHS) $\mathcal{H}_k$ of functions on a set $\mathcal{X}$. We assume that k is measurable and bounded, $\sup_{x \in \mathcal{X}} k(x, x) < \infty$. Then the MMD in $\mathcal{H}_k$ between two distributions P and Q over $\mathcal{X}$ is (Gretton et al., 2012a):

$$\text{MMD}_k^2(P, Q) := \mathbb{E}_{x,x'}\left[k(x, x')\right] + \mathbb{E}_{y,y'}\left[k(y, y')\right] - 2\,\mathbb{E}_{x,y}\left[k(x, y)\right] \tag{1}$$

where $x, x' \overset{iid}{\sim} P$ and $y, y' \overset{iid}{\sim} Q$. Many kernels, including the popular Gaussian RBF, are *characteristic* (Fukumizu et al., 2008; Sriperumbudur et al., 2010), which implies that the MMD is a metric, and in particular that $\text{MMD}_k(P, Q) = 0$ if and only if $P = Q$, so that tests with any characteristic kernel are consistent. That said, different characteristic kernels will yield different test powers for finite sample sizes, and so we wish to choose a kernel k to maximize the test power. Below, we will usually suppress explicit dependence on k.

Given $X = \{X_1, \dots, X_m\} \overset{iid}{\sim} P$ and $Y = \{Y_1, \dots, Y_m\} \overset{iid}{\sim} Q$,[2] one estimator of $\text{MMD}(P, Q)$ is

$$\widehat{\text{MMD}}_{\text{U}}^2(X, Y) := \frac{1}{\binom{m}{2}} \sum_{i \neq i'} k(X_i, X_{i'}) + \frac{1}{\binom{m}{2}} \sum_{j \neq j'} k(Y_j, Y_{j'}) - \frac{2}{\binom{m}{2}} \sum_{i \neq j} k(X_i, Y_j). \tag{2}$$

This estimator is unbiased, and has nearly minimal variance among unbiased estimators (Gretton et al., 2012a, Lemma 6).

Following Gretton et al. (2012a), we will conduct a hypothesis test with null hypothesis $H_0 : P = Q$ and alternative $H_1 : P \neq Q$, using test statistic $m\,\widehat{\text{MMD}}_{\text{U}}^2(X, Y)$. For a given allowable probability of false rejection α, we choose a test threshold c_α and reject H_0 if $m\,\widehat{\text{MMD}}_{\text{U}}^2(X, Y) > c_\alpha$.

Under $H_0 : P = Q$, $m\,\widehat{\text{MMD}}_{\text{U}}^2(X, Y)$ converges asymptotically to a distribution that depends on the unknown distribution P (Gretton et al., 2012a, Theorem 12); we thus cannot evaluate the test threshold c_α in closed form. We instead estimate a data-dependent threshold $\hat{c}_\alpha$ via permutation: randomly partition the data $X \cup Y$ into X' and Y' many times, evaluate $m\,\widehat{\text{MMD}}_{\text{U}}^2(X', Y')$ on each split, and estimate the $(1-\alpha)$th quantile $\hat{c}_\alpha$ from these samples. Section 3 discusses efficient computation of this process.

We now describe a mechanism to choose the kernel k so as to maximize the power of its associated test. First, note that under the alternative $H_1 : P \neq Q$, $\widehat{\text{MMD}}_{\text{U}}^2$ is asymptotically normal,

$$\frac{\widehat{\text{MMD}}_{\text{U}}^2(X, Y) - \text{MMD}^2(P, Q)}{\sqrt{V_m(P, Q)}} \xrightarrow{D} \mathcal{N}(0, 1), \tag{3}$$

where $V_m(P, Q)$ denotes the asymptotic variance of the $\widehat{\text{MMD}}_{\text{U}}^2$ estimator for samples of size m from P and Q (Serfling, 1980). The power of our test is thus, using Pr_1 to denote probability under H_1,

$$\begin{aligned}\text{Pr}_1\left(m\,\widehat{\text{MMD}}_{\text{U}}^2(X, Y) > \hat{c}_\alpha\right) &= \text{Pr}_1\left(\frac{\widehat{\text{MMD}}_{\text{U}}^2(X, Y) - \text{MMD}^2(P, Q)}{\sqrt{V_m(P, Q)}} > \frac{\hat{c}_\alpha/m - \text{MMD}^2(P, Q)}{\sqrt{V_m(P, Q)}}\right) \\ &\to \Phi\left(\frac{\text{MMD}^2(P, Q)}{\sqrt{V_m(P, Q)}} - \frac{c_\alpha}{m\sqrt{V_m(P, Q)}}\right)\end{aligned} \tag{4}$$

where Φ is the CDF of the standard normal distribution. The second step follows by (3) and the convergence of $\hat{c}_\alpha \to c_\alpha$ (Alba Fernández et al., 2008). Test power is therefore maximized by

[2] We assume for simplicity that the number of samples from the two distributions is equal.